



assembly or manufacturing units. Srinivas says that under the scheme, CIMEI has signed an MoU with the Uttar Pradesh state government, requesting 500 acres of land from the government to set up a complete electronics manufacturing ecosystem. As per the scheme, the government will bear the majority of the costs for land development and setting up a common facility centre.

“The production-linked incentive for the components is yet to be finalised. However, we have spoken to nearly 15 companies in Taiwan and China interested in setting up their manufacturing units as joint venture partners in India. Under this arrangement, the Indian partner will hold a 51% stake, and the foreign partner will hold 49%. This ensures that India has a major stake in the organisation, and manufacturing will occur within the country, thereby strengthening the supply chain and minimising disruptions, especially in component-level manufacturing,” adds Srinivas.

Research and development. One of the main reasons India’s neighbours accelerated their lead in the electronics revolution was the focus on developing technical capabilities.



“Passive components account for about 48% of the components, and active components comprise the remaining 52%. To foster domestic manufacturing, India can manufacture simple components such as transformers, capacitors, and transistors, which do not require extensive expertise.”

—Jairaj Srinivas, Director-General,
The Confederation of Indian MSMEs in ESDM and IT (CIMEI)

In contrast, India’s current spending on research and development in the MSME sector is only 2.5%. Small businesses often struggle to allocate funds for research and development due to limited financial resources. Despite having Centres of Excellence for different electronics domains and various incubation centres, the success rate for startups and small businesses working in the ESDM industry remains low.

To promote collaborative efforts with respect to research and development, Srinivas suggests collaboration and engagement by companies with substantial financial resources and significant corporate social responsibility (CSR) funds to invest in R&D initiatives. “These companies can contribute to the development of common facilities centres and establish contract research organisations that cater to various sectors, including the ESDM sector

and beyond. Small businesses can access research facilities without incurring substantial costs through such research opportunities, thereby promoting R&D activities,” he adds.

The electronics components market in India measures up to \$64.2 billion, of which a significant amount continues to be imported. To reduce dependency on imports and become truly *atmanirbhar* in the ESDM sector, initiatives such as the design-linked incentive scheme, for MSMEs and startups, propel manufacturers to increase domestic value addition, ultimately creating a high-value-addition electronics industry and generating employment in electronics component manufacturing. While the PLI schemes encourage investments in the sector, there is a dire need to establish and treat the components industry differently regarding incentives. Longer gestation period and a smaller investment-to-sales ratio call for extended incentive periods to ensure a positive return on investment in component manufacturing for investors. With nearly 50% of value addition in the component ecosystem, focusing on localised manufacturing can boost the nation’s overall economic growth and resilience.

While the Sony Walkman may be long gone and forgotten, we can certainly hope that soon enough every smartphone will have the majority, if not all, of the components completely manufactured, and not just assembled, in India. **EFY**

While manufacturing abilities may take some time to develop, harnessing what is in hand can help strengthen the sector. With many design houses and startups, incentivising design activity in India through the Design Linked Incentive (DLI) scheme will attract domestic and global players in the semiconductor industry. The scheme provides financial and infrastructural support to companies setting up fabs or semiconductor-making plants, encouraging innovation, reducing dependence on foreign manufacturers, and boosting India’s position in the digital revolution.

The DLI scheme consists of three components:

Chip design infrastructure support. C-DAC will establish the India Chip Centre to provide state-of-the-art design infrastructure and facilitate access to supported companies.

Product design linked incentive. Approved applicants engaged in semiconductor design can receive up to 50% reimbursement of eligible expenditures, with a ceiling of ₹150 million per application.

Deployment linked incentive. Approved applicants deploying semiconductor designs in electronic products can receive an incentive of 6% to 4% of net sales turnover over 5 years, with a ceiling of ₹300 million per application.

So far, seven startups with product applications ranging from artificial intelligence (AI) to vector processes and image sensors have been approved for funding. The scheme is part of the \$10 billion India semiconductor mission announced by the government in 2021.

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